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Inflammatory Breast Cancer and Racial Disparities in a SEER Database Analysis of Survival Outcomes

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Abstract Purpose Inflammatory breast cancer (IBC) has poor prognosis, and racial survival disparities have been reported. We formally evaluated whether the race–breast cancer-specific survival (BCSS) association differed by inflammatory status at comparable AJCC stages.

Methods We analyzed 3,174 women aged $\geq$ 20 years with AJCC 6th edition stage IIIB–IIIC breast cancer diagnosed in 2011–2015 in SEER. The cohort included 479 IBC and 2,695 non-IBC LABC cases. Cause-specific Cox models estimated adjusted associations with BCSS. A common Cox model with race-by-IBC interaction terms tested whether these associations differed by inflammatory status. Fine–Gray and propensity-score analyses were used as sensitivity analyses.

Results IBC was associated with a higher breast cancer-specific mortality hazard than non-IBC LABC (HR 1.415, 95% CI 1.196–1.675, $p < 0.001$). The adjusted Black-versus-White HR was 1.281 (95% CI 1.053–1.557, $p = 0.013$) within non-IBC LABC. Within IBC, it was 1.033 (95% CI 0.632–1.689, $p = 0.897$). The Black-by-IBC ratio of HRs was 0.807 (95% CI 0.477–1.365, $p = 0.423$). Neither the omnibus Wald test ($p = 0.538$) nor the likelihood-ratio test ($p = 0.528$) was statistically significant. The IBC comparison included 54 Black and 379 White women. Its approximate power to detect an HR of 1.281 was 20.17%.

Conclusion IBC was associated with a higher breast cancer-specific mortality hazard than non-IBC LABC. Black women also had a higher adjusted hazard than White women within non-IBC LABC. However, formal interaction testing did not provide statistically significant evidence that this association differed by inflammatory status. The interaction confidence interval was compatible with both smaller and larger associations within IBC. Larger cohorts with greater representation of Black patients with IBC are needed for more precise estimation.

Keywords inflammatory breast cancer · racial disparities · survival · SEER · competing risks · health equity

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1 Introduction

Inflammatory breast cancer (IBC) accounts for approximately 1%–5% of breast cancer diagnoses in the United States [2,26]. Earlier Surveillance, Epidemiology, and End Results (SEER) studies established its rarity and

described variation in incidence by age and race/ethnicity [1,14]. Clinically, IBC is characterized by rapid-onset diffuse breast erythema, edema, and peau d'orange, often without a discrete palpable mass [26,5,6]. Under the American Joint Committee on Cancer (AJCC) 6th edition staging system, inflammatory carcinoma is classified as T4d disease [23]. The T4d designation was retained in the AJCC 7th edition [24]. Patients frequently present with extensive nodal involvement or distant metastases, and outcomes remain poorer than those of patients with non-inflammatory breast cancer despite multimodal treatment with systemic therapy, surgery, and radiation [6,7,8,9,10,34]. Breast cancer broadly comprises molecularly distinct subtypes [11]. Within IBC specifically, prognosis varies according to hormone receptor (HR) and human epidermal growth factor receptor 2 (HER2) status, and substantial proportions of IBC are HER2-positive or HR-negative/HER2-negative [10,12,15].

Evidence from broader advanced breast cancer populations further demonstrates the prognostic importance of HER2 status and trastuzumab treatment, while triple-negative breast cancer remains clinically challenging because it lacks established hormone receptor- or HER2-directed treatment targets [16,17]. More broadly, contemporary breast cancer treatment frameworks use HR and HER2 status to guide the escalation or de-escalation of systemic treatment [13]. Survival differences among patients with aggressive breast cancer may reflect a complex combination of tumor characteristics, treatment access and completion, comorbidity, socioeconomic conditions, and other healthcare-related factors [10,18,32,37].

Racial disparities in breast cancer survival are well documented in population-based studies. Breast cancer incidence patterns also vary according to age and race/ethnicity [3]. Separately, Non-Hispanic Black women experience higher breast cancer mortality than Non-Hispanic White women [4,18]. Potential contributors include differences in stage at diagnosis, tumor characteristics, treatment receipt and completion, healthcare access, comorbidity, socioeconomic conditions, and structural inequities [10,18,32,37]. Registry- and population-based studies have examined racial and ethnic differences in IBC incidence, treatment, and survival across several populations and diagnosis periods [32,36,37,38,39]. A recent systematic review and meta-analysis reported higher mortality among Non-Hispanic Black patients with IBC than among Non-Hispanic White patients while emphasizing heterogeneity and residual confounding across the available observational studies [40].

Although racial disparities in IBC have been studied, estimates vary across study populations, diagnosis periods, case definitions, survival endpoints, and covariate adjustment sets [32,36,37,38,39,40]. The specific question of whether the association between race and breast cancer-specific survival (BCSS) differs between IBC and non-IBC locally advanced breast cancer (LABC) at comparable AJCC stages remains unresolved. Answering this question requires both inflammatory-status-specific estimates and a formal race-by-IBC interaction test. Differences in statistical significance between subgroup-specific estimates alone do not establish that the race-survival association differs by inflammatory status.

Outcome definition is also important. Overall survival combines breast cancer deaths with deaths from other causes, which may obscure differences in breast cancer-specific mortality, particularly in older populations. Cause-specific Cox models estimate associations with the rate of breast cancer death among patients who remain alive and event-free. Fine–Gray models estimate covariate associations with the subdistribution hazard of breast cancer death in the presence of competing mortality and provide inference about cumulative incidence [19,20,21]. Evaluating both estimands provides complementary information about race-associated differences in breast cancer mortality.

To address these gaps, we conducted a retrospective cohort study using the Surveillance, Epidemiology, and End Results (SEER) database to evaluate BCSS among women with AJCC 6th edition stage IIIB–IIIC breast cancer diagnosed from 2011 through 2015. Restricting the cohort to stage IIIB–IIIC improved clin-

ical comparability between IBC and non-IBC LABC. The primary objective was to estimate race associations within each inflammatory-status group and formally test whether the race-BCSS association differed by inflammatory status using race-by-IBC interaction terms. Secondary objectives were to assess the findings using cause-specific Cox and Fine-Gray competing-risk models and to evaluate sensitivity to measured baseline differences using propensity-score methods. Because the data are observational, all effect estimates are interpreted as associations. The analyses cannot establish the biological, treatment-related, healthcare, or social mechanisms underlying racial survival differences, nor can they establish that inflammatory status causally modifies those differences.

The rest of the paper is organized as follows. Section 2 describes the methodology of the study. It presents the data sources, data preparation steps, data processing flowchart and the statistical analyses used in the paper. Section 3 reports the results of all statistical analyses and Section 4 discusses the main findings and their implications. Section 5 provides the concluding remarks.

2 Methods

2.1 Study population and data source

This retrospective cohort study used data from the Surveillance, Epidemiology, and End Results (SEER) program, a population-based cancer registry maintained by the National Cancer Institute [22]. The source cohort contained 87,915 women aged 20 years or older who were diagnosed with invasive breast cancer between 2011 and 2015. Eligibility was restricted to American Joint Committee on Cancer (AJCC) 6th edition stage IIIB or IIIC disease. AJCC 6th edition staging was used because SEER provides a standardized derived AJCC 6th edition stage variable for these diagnosis years.

Among patients with stage IIIB-IIIC disease, inflammatory breast cancer (IBC) was defined as recorded T4d disease according to the AJCC 6th edition staging system [23]. The T4d designation was retained in the AJCC 7th edition [24]. Patients without a recorded T4d classification were categorized as having non-inflammatory locally advanced breast cancer (LABC).

Of the 87,915 records in the source cohort, 84,555 were excluded because they did not meet the stage IIIB–IIIC eligibility criterion, leaving 3,360 stage-eligible patients. We then excluded 139 patients with unknown HR/HER2-defined molecular subtype, seven patients with indeterminate SEER Surgery of Primary Site codes, including surgery not otherwise specified (code 90, $n = 1$) and unknown whether surgery was performed or death-certificate-only identification (code 99, $n = 6$), and 40 patients whose race was recorded as American Indian or Alaska Native ($n = 33$) or unknown ($n = 7$).

The final analytic cohort comprised 3,174 women, including 479 patients with IBC (15.1%) and 2,695 patients with non-inflammatory LABC (84.9%). The 186 post-stage-eligibility exclusions represented 5.5% of the 3,360 stage-eligible patients. Figure 1 summarizes the cohort-selection and analysis workflow.

2.2 Data preparation and variable creation

Survival time was calculated in months from the date of diagnosis to the date of death or last contact. For patients whose diagnosis and death occurred during the same calendar month, survival time was set to 0.5 months, corresponding to approximately 15 days. This avoided zero-valued survival times. SEER documentation indicates that survival shorter than one month may be recorded as zero months [26], while Woods et al. evaluated assigning 15 days to such observations [25]. In their breast cancer analysis, this assignment underestimated 1-year relative survival by 0.2 percentage points compared with estimates based on complete dates [25].

Age at diagnosis was categorized as 49 years or younger and older than 49 years. These categories were used as age-based proxies for premenopausal and postmenopausal status, respectively, rather than as directly observed menopausal status [35]. Race was obtained from the SEER race variable and categorized as White, Black, or Asian or Pacific Islander (API). Hispanic ethnicity was not used to define or exclude these analytic groups.

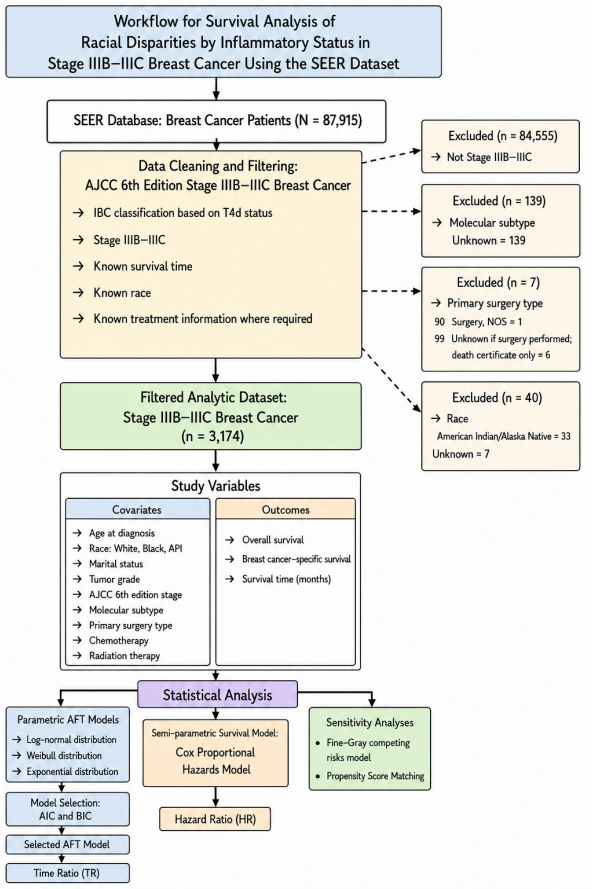


Fig. 1: Data processing flowchart

American Indian or Alaska Native patients ($n = 33$) were excluded because the sample was too small for stable group-specific estimates. Patients with unknown race ($n = 7$) were excluded because they could not be assigned to an analytic race category.

Clinical and pathologic variables included tumor grade, AJCC 6th edition stage, and molecular subtype. Molecular subtype was constructed from hormone receptor (HR) and human epidermal growth factor receptor 2 (HER2) status and categorized as Luminal A, Luminal B, HER2 enriched, or triple-negative breast cancer.

Only two patients had Grade IV tumors. Grade III and Grade IV represent poorly differentiated and undifferentiated high-grade disease, respectively. Modeling Grade IV as a separate regression category with only two observations would create an extremely sparse model level and could produce unstable coefficient and variance estimates. Grades III and IV were therefore combined into a single high-grade category to retain adjustment for tumor differentiation while avoiding estimation based on a two-patient category. This modeling decision does not imply prognostic equivalence between Grade III and Grade IV disease. Residual prognostic heterogeneity between the two grades may remain and is explicitly acknowledged as a study limitation. As an additional sensitivity analysis, the primary race-by-IBC interaction model was refitted using the same model specification after excluding the two patients with Grade IV disease to assess whether the pooling decision materially influenced the primary interaction results.

Treatment variables included primary surgery type, chemotherapy, and radiation therapy. Primary surgery was categorized as segmental or breast-conserving surgery, mastectomy, or no surgery. The complete variable definitions and the abbreviated labels used in subsequent tables and figures are presented in Table 1.

2.3 Outcome measures

The primary outcome was breast cancer-specific survival (BCSS), defined as the interval from diagnosis to death attributed to breast cancer according to the SEER cause-specific death classification [26]. Patients who were alive at last follow-up were censored at the date of last contact, while patients who died from causes other than breast cancer were censored at the date of death. The BCSS analysis included 1,187 breast cancer-specific deaths and 1,987 censored observations.

Overall survival (OS) was treated as a secondary outcome and was defined as the time from breast cancer diagnosis to death from any cause. Patients who remained alive were censored at the date of last contact. For competing-risk analyses, breast cancer death was treated as the event of interest, death from another cause was treated as a competing event, and patients who remained alive were censored at the date of last contact.

BCSS was selected as the primary outcome because it focuses on mortality attributed to breast cancer, whereas

Table 1: Demographic, clinical, pathologic, and treatment variables included in the study

Variable	Categories
Age at diagnosis	Pre-menopausal: $\leq$ 49 years Post-menopausal: $>$ 49 years
Race	White Black API (Asian or Pacific Islander)
Marital status	Married Unmarried Unknown
Tumor grade	Grade I: Well differentiated Grade II: Moderately differentiated Grade III or IV: Poorly differentiated or undifferentiated
AJCC 6th edition stage	Stage IIIB Stage IIIC
Molecular subtype	HR+ and HER2–: Luminal A HR+ and HER2+: Luminal B HR– and HER2+: HER2 enriched HR– and HER2–: Triple-negative breast cancer
Primary surgery type	Segmental or breast-conserving surgery Mastectomy No surgery
Chemotherapy	Yes No
Radiation therapy	Yes No Unknown

OS also reflects mortality from unrelated causes. Other-cause mortality was evaluated as a secondary outcome. The statistical methods used to analyze these outcomes are described in the next section.

2.4 Statistical Analysis

Baseline demographic, clinicopathologic, and treatment characteristics were summarized for the overall cohort and by inflammatory breast cancer status. Categorical variables were reported as counts and column percentages, with percentages calculated separately within the IBC and non-IBC LABC groups. Differences between groups were evaluated using Pearson's chi-squared tests. Survival time was summarized using the median, first and third quartiles, and range.

Breast cancer-specific survival (BCSS) was estimated using the Kaplan–Meier product-limit estimator [27]. Patients who died from causes other than breast cancer were censored at the time of death, as defined in Section 2.3. Kaplan–Meier curves were constructed by IBC status, race, and HR/HER2-defined molecular subtype. Differences in survival distributions were evaluated using log-rank tests. Three-year BCSS probabilities, median BCSS, and corresponding 95% confidence intervals (CIs) were estimated. Median BCSS was reported as not reached when the estimated survival probability remained above 0.50 throughout follow-up.

The primary multivariable analysis used a cause-specific Cox proportional hazards model to estimate associations with breast cancer mortality [28]. The model included IBC status, race, age at diagnosis, molecular subtype, tumor grade, AJCC 6th edition stage, marital status, primary surgery type, chemotherapy, and radiation therapy. The reference categories were non-IBC LABC, White race, age 49 years or younger, HR+/HER2– disease, Grade I, stage IIIB, married status, segmental or breast-conserving surgery, no or unknown chemotherapy, and no radiation therapy. Results were reported as hazard ratios (HRs), 95% CIs, and two-sided p-values. The Efron method was used to handle tied event times.

The proportional hazards assumption was evaluated using tests based on scaled Schoenfeld residuals and graphical inspection of scaled residuals against time [43]. A secondary stratified Cox model allowed the baseline hazard to vary across molecular subtype, tumor grade, primary surgery type, and radiation therapy strata. IBC status, race, age, AJCC stage, marital status, and chemotherapy remained regression covariates in the stratified model. In the initial multivariable Cox model, the overall tumor-grade term showed evidence of nonproportional hazards based on scaled Schoenfeld residuals. Tumor grade was therefore included as a stratification factor in the stratified sensitivity model, allowing separate baseline hazards across Grade I, Grade II, and combined Grade III/IV categories and avoiding a common proportional-hazards constraint for grade.

To address the primary research question, a common stratified Cox model was fitted to formally test whether the association between race and BCSS differed by inflammatory status [28,42]. Black and API indicator variables were created using White race as the reference category. The model included Black-by-IBC and API-by-IBC interaction terms and adjusted for age, AJCC stage, marital status, and chemotherapy. The baseline hazard was stratified by IBC status, molecular subtype, tumor grade, primary surgery type, and radiation therapy. Because IBC status was used as a stratification variable, this interaction model did not estimate a single main-effect HR for IBC status.

Black-versus-White and API-versus-White HRs within the IBC and non-IBC LABC groups were obtained as linear contrasts from the common interaction model. Each interaction estimate represented the IBC-to-non-IBC ratio of the corresponding race-specific HR. Individual interaction terms were evaluated using Wald tests. The two interaction terms were jointly evaluated using a two-degree-of-freedom omnibus Wald test and a likelihood-ratio test comparing models with and without the interaction terms. Evidence that the race–BCSS association differed by inflammatory status was based on these formal interaction tests. A statistically significant association in one subgroup and a nonsignificant association in another subgroup was not interpreted as evidence of interaction [42]. The proportional hazards assumption for the interaction model was evaluated using scaled Schoenfeld residual tests [43].

Because the Black IBC subgroup was small, statistical precision was evaluated using an event-based minimum detectable effect calculation based on the Schoenfeld proportional-hazards approximation [44]. The calculation used the observed number of breast cancer-specific deaths among Black and White patients with IBC, the observed proportion of Black patients, a two-sided significance level of 0.05, and 80% power. We calculated the minimum detectable HR above 1 and its reciprocal below 1. We also calculated the approximate power to detect an association equal in magnitude to the adjusted non-IBC Black-versus-White HR. This retrospective calculation was used to characterize the precision of the IBC subgroup analysis and was not interpreted as prospective study power.

Race associations were also estimated separately within each molecular subtype using adjusted Cox models. Because molecular subtype was constant within each analysis, it was not included as a covariate in these models. Each subtype-specific model included IBC status, race, age, AJCC stage, marital status, primary surgery type, chemotherapy, tumor grade, and radiation therapy when the corresponding variable was not used for stratification. The triple-negative breast cancer model was stratified by tumor grade and radiation therapy. The Luminal A model was stratified by tumor grade and primary surgery type. The Luminal B model was stratified by tumor grade. The HER2-enriched model was stratified by radiation therapy. The proportional hazards assumption was evaluated separately for each subtype-specific model using scaled Schoenfeld residuals.

In a separate exploratory analysis, a common Cox model containing race, molecular subtype, and race-by-molecular-subtype interaction terms was fitted to formally assess whether the association between race and BCSS differed across molecular subtypes [28,42]. White race and Luminal A disease served as the reference categories. The model adjusted for age group, AJCC 6th edition stage, marital status, and chemotherapy. Baseline hazards were stratified by IBC status, tumor grade, primary surgery type, and radiation therapy. Each interaction estimate represented the ratio of the race-specific HR in the corresponding molecular subtype to the race-specific HR in Luminal A disease. The six race-by-molecular-subtype interaction terms were evaluated individually using Wald tests and jointly using a six-degree-of-freedom omnibus Wald test. A likelihood-ratio test also compared models with and without the interaction terms. Differences in the magnitudes or statistical significance of the separate subtype-specific estimates were not interpreted as evidence of effect modification without support from the formal interaction tests.

Accelerated failure time models assuming log-normal, Weibull, and exponential survival distributions were fitted to evaluate sensitivity to alternative survival-model specifications [29]. Fully adjusted AFT models included IBC status, race, age, molecular subtype, tumor grade, AJCC stage, marital status, primary surgery type, chemotherapy, and radiation therapy. Model coefficients were exponentiated and reported as time ratios with 95% CIs obtained from the corresponding model-based standard errors. A time ratio below 1 indicated shorter survival relative to the reference category, while a time ratio above 1 indicated longer survival. Distributional fit was compared using the Akaike information criterion and Bayesian information criterion, with lower values indicating better relative fit among the candidate models.

Overall survival was evaluated as a secondary outcome using a multivariable Cox proportional hazards model. Death from any cause was treated as the event, and patients alive at last contact were censored. The overall-survival model adjusted for age, race, molecular subtype, tumor grade, AJCC stage, primary surgery type, chemotherapy, radiation therapy, and marital status.

Competing-risk analyses treated breast cancer death as the event of interest, death from another cause as a competing event, and patients alive at last contact as censored. Cumulative incidence functions for breast cancer death and other-cause death were estimated by IBC status using the Aalen–Johansen approach [20, 21]. A multivariable Fine–Gray subdistribution hazards model was fitted as a sensitivity analysis [19]. The Fine–Gray model included IBC status, race, age, molecular subtype, tumor grade, AJCC stage, marital status, primary surgery type, chemotherapy, and radiation therapy. Results were reported as subdistribution hazard ratios with 95% CIs. Cause-specific Cox and Fine–Gray estimates were interpreted as complementary because they address different aspects of breast cancer mortality in the presence of competing events [19, 20, 21].

A separate propensity-score sensitivity analysis was conducted among White and Black women. Propensity scores for Black race were estimated using logistic regression containing age at diagnosis, year of diagnosis, AJCC stage, tumor grade, HR/HER2-defined molecular subtype, primary surgery type, chemotherapy, radiation therapy, and marital status [30, 31]. Matching was performed separately within the IBC and non-IBC cohorts using `MatchIt` [41]. Greedy 1:1 nearest-neighbor matching was performed without replacement. Black women were designated as the focal group, and the `MatchIt` ATT setting was used to define the matched comparison. Focal-group patients were processed in descending order of the estimated propensity score. The implementation specified `distance = "glm"`, `link = "logit"`, `estimand = "ATT"`, `ratio = 1`, `replace = FALSE`, `m.order = "largest"`, `discard = "none"`, `caliper = 0.20`, and `std.caliper = TRUE`. Thus, with `std.caliper = TRUE`, the exact caliper width was 0.20 standard deviations of the estimated propensity-score distance. No exact-matching restrictions were imposed, and no additional common-support discard rule was applied beyond the caliper. Because race is not a manipulable treatment, the matched estimates were interpreted as adjusted associations rather than causal treatment effects.

Covariate balance was assessed before and after matching using absolute standardized mean differences (SMDs). SMDs were used because they quantify the magnitude of between-group covariate imbalance on a common standardized scale and, unlike hypothesis-test p-values, are not driven by sample size. A matched sample was considered to have acceptable measured balance only when every included baseline covariate had $|\text{SMD}| < 0.10$. The 0.20 caliper governed the maximum allowable difference in the estimated propensity-score distance between matched patients, whereas the absolute SMD threshold of 0.10 was used separately to evaluate covariate balance after matching. The caliper did not guarantee that every measured covariate would satisfy the balance criterion. We selected 0.10 as a conventional diagnostic threshold because it is widely used in propensity-score analyses to identify potentially meaningful residual covariate imbalance [31]. The same criterion was applied uniformly to every included covariate

in both the IBC and non-IBC LABC matched samples. This threshold was treated as a practical balance diagnostic rather than evidence that residual or unmeasured confounding had been eliminated. Balance was evaluated separately within the two matched samples and displayed using Love plots. A matched sample containing any covariate with an absolute SMD of 0.10 or greater was considered to retain residual measured imbalance and was not described as adequately balanced.

Post-matching Black-versus-White associations with BCSS were estimated separately within the IBC and non-IBC LABC matched samples using Cox proportional hazards models with robust sandwich standard errors. The matched analyses were treated as sensitivity analyses addressing measured baseline differences rather than as causal effect estimates. Matching cannot account for unmeasured confounding, and the IBC matched sample retained measured imbalance and limited precision. Accordingly, the matched IBC estimate was interpreted as inconclusive rather than as evidence for the presence or absence of a racial survival association.

All coefficient-level Wald tests were two-sided because associations in either direction were scientifically plausible. Omnibus Wald and likelihood-ratio tests used chi-squared reference distributions. A nominal significance level of $\alpha = 0.05$ was used for the primary race-by-IBC effect-modification analysis and applied to the individual Wald tests, omnibus Wald test, and likelihood-ratio test. The same nominal threshold was used for secondary formal tests. Analyses were conducted using R version 4.5.1 [33], including `survival` version 3.8.3, `cmprsk` version 2.2.12, `MatchIt` version 4.7.2, `cobalt` version 4.6.3, `gtsummary` version 2.4.0, and `survminer` version 0.5.1.

3 Results

The results of the study are presented below. We first summarize baseline demographic, clinical, pathologic, and treatment characteristics of the study cohort, stratified by inflammatory breast cancer status. Subsequent sections present survival analyses and examine racial disparities in breast cancer-specific survival by inflammatory breast cancer status.

3.1 Baseline Characteristics

The final analytic cohort included 3,174 women diagnosed with Stage IIIB or IIIC breast cancer. Baseline demographic and clinical characteristics stratified by IBC status are summarized in Table 2. IBC patients (n = 479, 15.10%) were significantly more likely to present with Grade III/IV tumors, triple-negative molecular subtype, and AJCC Stage IIIB disease compared to non-IBC LABC patients (n = 2,695, 84.90%). Racial distributions were comparable between IBC and non-IBC LABC groups ($p = 0.20$) and these were similar across groups: among non-IBC LABC patients, 76.25% were White, 11.47% were Black, and 12.28% were API, while among IBC patients these proportions were 79.13% White, 11.27% Black, and 9.60% API. Treatment patterns also differed, with IBC patients more frequently receiving mastectomy (81.21%), chemotherapy (89.56%), and radiation therapy (71.82%). These differences describe recorded treatment patterns and do not establish treatment appropriateness, timing, or guideline concordance. Only two Grade IV tumors were observed; therefore, Grades III and IV were combined and are presented as Grade III/IV in all primary analyses.

3.2 Survival Analysis

We evaluated breast cancer-specific survival (BCSS) in the overall cohort and stratified by demographic and clinical variables. In the overall cohort, the 3-year BCSS was 80.5% (95% CI 79.1–82.0). Substantial racial disparities in survival were evident across race groups, shown in Table S1. White women had 3-year BCSS of 81.2% (95% CI 79.6–82.7), Black women 70.9% (95% CI 66.4–75.7), and API women 85.6% (95% CI 82.3–89.0). Black women experienced substantially worse survival (median 109 months) compared to White and API (median not reached), with these racial differences persisting throughout follow-up as shown in Figure S1. These

Table 2: Baseline characteristics of the study population by inflammatory breast cancer (IBC) status

Characteristic	Non-IBC LABC (N = 2,695)[1]	IBC (N = 479)[1]	p-value[2]
Race			0.200
White	2,055 (76.25%)	379 (79.13%)	
Black	309 (11.47%)	54 (11.27%)	
API	331 (12.28%)	46 (9.60%)	
Age			< 0.001
Pre-menopausal	593 (22.00%)	154 (32.15%)	
Post-menopausal	2,102 (78.00%)	325 (67.85%)	
Molecular subtype			< 0.001
HR+/HER2-	1,697 (62.97%)	185 (38.62%)	
HR+/HER2+	348 (12.91%)	100 (20.88%)	
HR-/HER2+	221 (8.20%)	78 (16.28%)	
HR-/HER2-	429 (15.92%)	116 (24.22%)	
Tumor grade			< 0.001
Grade I	229 (8.50%)	13 (2.71%)	
Grade II	1,160 (43.04%)	173 (36.12%)	
Grade III/IV	1,306 (48.46%)	293 (61.17%)	
AJCC stage			< 0.001
IIIB	1,022 (37.92%)	387 (80.79%)	
IIIC	1,673 (62.08%)	92 (19.21%)	
Marital status			0.020
Married	2,129 (79.00%)	351 (73.28%)	
Unmarried	465 (17.25%)	106 (22.13%)	
Unknown	101 (3.75%)	22 (4.59%)	
Primary surgery type			< 0.001
Segmental	446 (16.55%)	11 (2.30%)	
Mastectomy	1,975 (73.28%)	389 (81.21%)	
No surgery	274 (10.17%)	79 (16.49%)	
Chemotherapy			< 0.001
No	672 (24.94%)	50 (10.44%)	
Yes	2,023 (75.06%)	429 (89.56%)	
Radiation therapy			0.012
No	825 (30.61%)	120 (25.05%)	
Yes	1,749 (64.90%)	344 (71.82%)	
Unknown	121 (4.49%)	15 (3.13%)	

[1]n (column %)
[2]Pearson's Chi-squared test

are unadjusted descriptive comparisons across the race categories used in SEER.

Given the marked survival differences between inflammatory and non-inflammatory disease, we next examined whether racial disparities in survival differed by inflammatory breast cancer status. Survival distributions differed significantly by IBC status and molecular subtype (Figure 2). Patients with IBC had significantly worse outcomes compared to non-IBC LABC patients (Figure 2a). Molecular subtype was a strong prognostic factor (Figure 2b), with the best survival observed in the HR+/HER2+ (Luminal B) subtype and the poorest survival in the triple-negative subtype.

Racial disparities were also evident, with Black women experiencing significantly worse survival compared to White and API women. For TNBC patients, the 3-year BCSS was 56.5% (95% CI 52.4– 60.9), and the median survival was only 52 months. Race-stratified breast cancer-specific survival within the TNBC subgroup is shown in Figure S2.

To evaluate the robustness of survival estimates to alternative distributional assumptions, three accelerated failure time (AFT) models assuming log-normal, Weibull, and exponential distributions were fitted to the observed breast cancer-specific survival data. Model fit was compared using Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), with lower values indicating superior fit. The log-normal distribution demonstrated the best model fit (AIC = 14,699.36; BIC = 14,729.67), compared to the Weibull (AIC = 14,813.71; BIC = 14,844.02) and exponential (AIC = 14,812.75; BIC = 14,837.00) models (Table S3). In log-normal, Weibull, and exponential accelerated failure time (AFT) models, a time ratio greater than 1 indicates longer survival compared with the reference group, whereas a time ratio less than 1 indicates shorter survival. Results of AFT models showed that Black women exhibited shorter estimated breast cancer-specific survival times relative to White women (time ratios = 0.844, 0.813, 0.808, respectively), whereas API women exhibited longer estimated survival times relative to White women (time ratios = 1.274, 1.213, 1.225, respectively). The direction of the race coefficients (Black time ratio range: 0.808–0.844; API time ratio range: 1.213–1.274) was similar across the three fitted AFT models, although statistical evidence varied by model and distributional choice (Table S5). Treatment coefficients capture associations rather than true causal effects of treatment, since the choice of treatment may depend on disease severity, eligibility criteria, and other unmeasured factors.

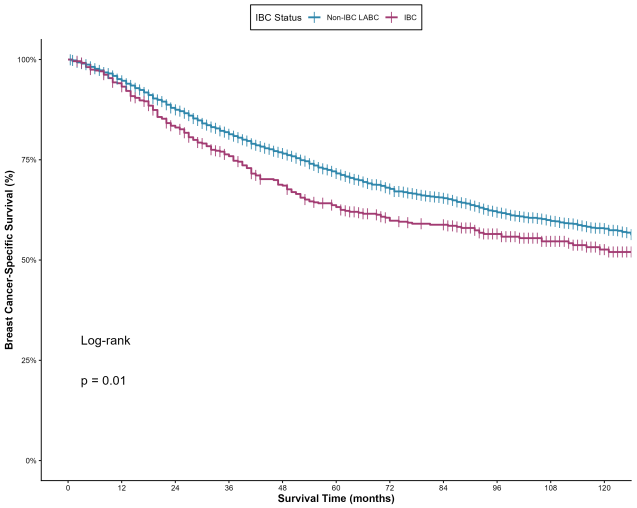
Associations for treatment and tumor characteristics were generally consistent across the fitted AFT models. Segmental surgery was associated with longer estimated survival than mastectomy or no surgery, while chemotherapy and radiation were also associated with longer estimated survival. These treatment coefficients

should not be interpreted as causal effects because treatment selection may depend on disease severity, eligibility, timing, and other measured or unmeasured clinical factors. Higher tumor grade and advanced AJCC stage were associated with shorter survival, while married women had longer estimated survival than unmarried women and those with unknown marital status (Table S5). Overall, the log-normal AFT model provided the best fit, while the direction of the race coefficients was consistent across the log-normal, Weibull, and exponential models (Tables S3 and S5).

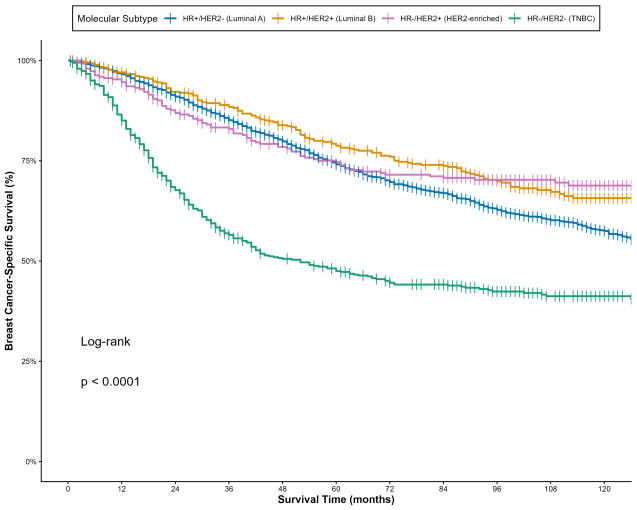
3.3 Multivariable Analysis

The results of the multivariable Cox proportional hazards model are visualized in the forest plot in Figure 3. After adjustment for the measured covariates, IBC was associated with a higher breast cancer-specific mortality hazard (HR 1.415, 95% CI 1.196–1.675, $p < 0.001$). Black race was associated with a 23.5% higher hazard of death compared to White race (HR 1.235, 95% CI 1.037–1.472, $p = 0.018$). In contrast, API women had an 18.7% lower breast cancer-specific mortality hazard than White women (HR 0.813, 95% CI 0.671–0.985, $p = 0.034$). These estimates are conditional associations and do not exclude residual confounding by unmeasured tumor, treatment, comorbidity, socioeconomic, or healthcare-access factors. Triple-negative breast cancer was associated with the highest mortality hazard among molecular subtypes (HR 1.998, 95% CI 1.715–2.327, $p < 0.001$) relative to the Luminal A subtype. Receipt of chemotherapy and radiation therapy was associated with lower mortality hazards, but these coefficients should not be interpreted as causal treatment effects.

Although race was associated with breast cancer-specific mortality in the overall adjusted model, this estimate represents an average effect across inflammatory and non-inflammatory disease and does not assess whether the association differs by inflammatory breast cancer status.



(a) IBC vs. Non-IBC



(b) Molecular Subtype

Fig. 2: Kaplan–Meier estimates of breast cancer-specific survival comparing (a) inflammatory breast cancer (IBC) vs. non-inflammatory locally advanced breast cancer (non-IBC LABC), and (b) molecular subtype

3.4 Racial Disparities Stratified by Inflammatory Breast Cancer Status

To address the primary research question, we fitted a common stratified multivariable Cox proportional hazards model containing race and race-by-IBC interaction terms, with the baseline hazard stratified by IBC status. Among patients with non-IBC LABC, Black women had a higher adjusted breast cancer-specific mortality hazard than White women (HR 1.281, 95% CI 1.053–1.557, p=0.013). The API-versus-White estimate within

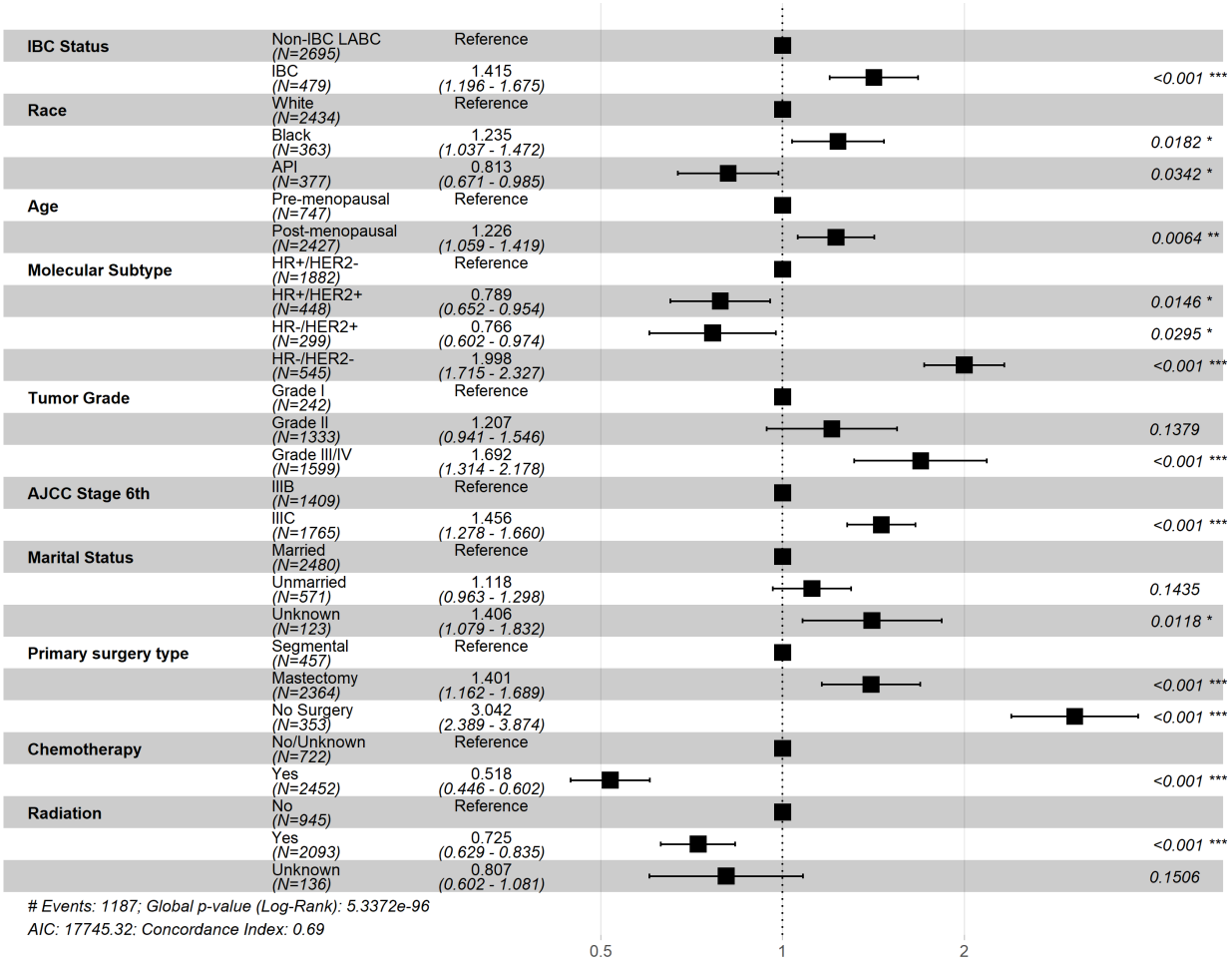


Fig. 3: Forest Plot of Multivariable Cox Proportional Hazards Model.

non-IBC LABC was HR 0.835 (95% CI 0.676–1.030, p=0.093). Within the IBC subgroup, the Black-versus-White estimate was HR 1.033 (95% CI 0.632–1.689, p=0.897), and the API-versus-White estimate was HR 0.640 (95% CI 0.356–1.149, p=0.135). These adjusted estimates are presented in Table 3, and the corresponding unadjusted Kaplan–Meier curves are shown in Figure 4.

The Black-by-IBC interaction estimate, expressed as the IBC-to-non-IBC ratio of the Black-versus-White HRs, was 0.807 (95% CI: 0.477–1.365, $p = 0.423$). The corresponding API-by-IBC interaction estimate was 0.766 (95% CI: 0.411–1.427, $p = 0.401$). The omnibus Wald test of the race-by-IBC interaction terms was not statistically significant ($\chi^2(2) = 1.239$, $p = 0.538$). The likelihood-ratio test comparing models with and without the race-by-IBC interaction terms was also not statistically significant ($\chi^2(2) = 1.278$, $p = 0.528$). Therefore, the analysis did not demonstrate that the race-BCSS association differed by inflammatory status. The statistically significant Black-versus-White association within non-IBC LABC and the nonsignificant association within IBC do not themselves establish effect modification. The Black-by-IBC ratio of HRs was below 1, but its 95% confidence interval included 1 and spanned values compatible with both a smaller and a larger Black-versus-White association within IBC relative to non-IBC LABC. The data therefore do not establish the direction or magnitude of race-by-inflammatory-status effect modification.

The event-based precision analysis included 54 Black women and 379 White women with IBC, among whom

188 breast cancer-specific deaths were observed. Given this subgroup composition, the minimum detectable HR with 80% power was 1.856, with a reciprocal detectable HR of approximately 0.539. The approximate power to detect an HR of 1.281, corresponding to the observed Black-versus-White estimate within non-IBC LABC, was only 20.17% (Table S7). These results indicate that the IBC subgroup had insufficient precision to reliably detect an association comparable in magnitude to that observed within non-IBC LABC. Accordingly, the non-significant Black-versus-White estimate within IBC should not be interpreted as evidence that no racial survival association existed. As an additional sensitivity analysis, the primary race-by-IBC interaction model was refitted after excluding the two patients with Grade IV disease. The Black-by-IBC ratio of HRs was 0.800 (95% CI 0.473–1.354, $p = 0.406$). Neither the omnibus Wald test ($\chi^2(2) = 1.284$, $p = 0.526$) nor the likelihood-ratio test comparing models with and without the race-by-IBC interaction terms ($\chi^2(2) = 1.325$, $p = 0.516$) was statistically significant. These estimates were very similar to those from the primary analysis, indicating that exclusion of the two Grade IV cases did not materially alter the race-by-IBC interaction estimate or its interpretation.

Exploratory subtype-specific Cox models were also used to estimate race associations within HR/HER2-defined molecular subtypes, as presented in Table S4. Among patients with triple-negative breast cancer, the Black-versus-White estimate was HR 1.334 (95% CI 0.977–1.821, p=0.070). Among patients with Luminal A disease, the Black-versus-White estimate was HR 1.129 (95% CI 0.876–1.456, p=0.347), whereas API women had a lower estimated breast cancer-specific mortality hazard than White women (HR 0.704, 95% CI 0.539–0.920, p=0.010).

A common Cox model containing race-by-molecular-subtype interaction terms was fitted to formally assess whether the association between race and BCSS differed across molecular subtypes. The individual interaction estimates are presented in Table S8. None of the individual interaction terms reached statistical significance. The omnibus race-by-molecular-subtype interaction was not statistically significant by either the Wald test ($\chi^2(6) = 6.145$, $p = 0.407$) or the likelihood-ratio test ($\chi^2(6) = 5.998$, $p = 0.423$). Therefore, the analysis did not demonstrate that molecular subtype modified the association between race and BCSS. Differences in the magnitudes or statistical significance of the subtype-specific estimates in Table S4 should not be interpreted as evidence of effect modification. The nominally significant API-versus-White association within Luminal A disease should also be interpreted as exploratory because the omnibus interaction was not statistically significant and multiple subtype-specific comparisons were examined.

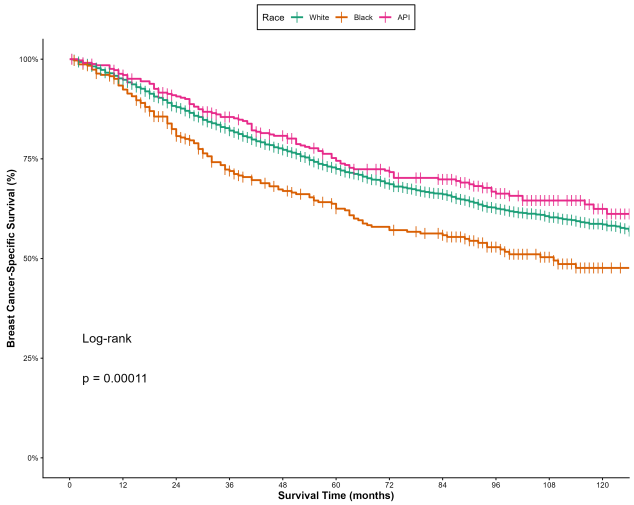
Table 3: Adjusted race associations with breast cancer-specific survival by inflammatory breast cancer status

Comparison	HR	95% CI	*p*-value
Non-IBC LABC			
Black vs. White (Ref)	1.281	(1.053–1.557)	0.013
API vs. White (Ref)	0.835	(0.676–1.030)	0.093
Inflammatory breast cancer			
Black vs. White (Ref)	1.033	(0.632–1.689)	0.897
API vs. White (Ref)	0.640	(0.356–1.149)	0.135
Race-by-IBC interaction			
Black-by-IBC interaction	0.807	(0.477–1.365)	0.423
API-by-IBC interaction	0.766	(0.411–1.427)	0.401

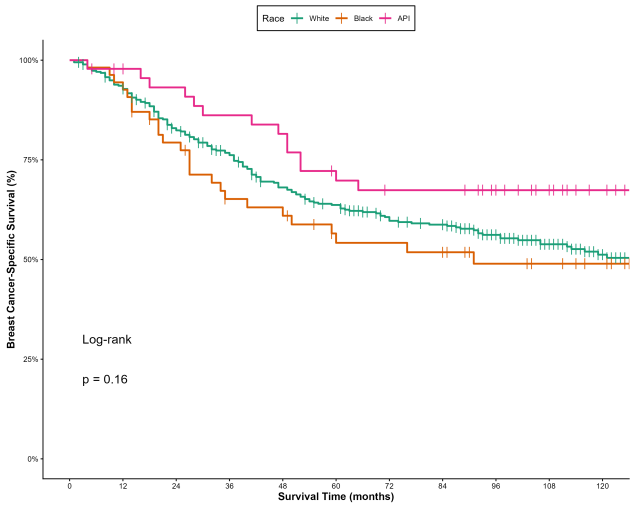
HRs compare each racial group with White women as the reference group. Interaction estimates are IBC-to-non-IBC ratios of HRs. Omnibus interaction: Wald $\chi^2(2) = 1.239$, p=0.538; likelihood-ratio $\chi^2(2) = 1.278$, p=0.528

3.4.1 Model Diagnostics

Scaled Schoenfeld-residual testing of the initial multivariable Cox model identified nonproportionality for the overall tumor-grade term ($\chi^2 = 93.095$, df = 2, $p < 0.001$). In the stratified sensitivity model, tumor grade, molecular subtype, primary surgery type, and radiation therapy were treated as stratification factors, thereby relaxing the proportional-hazards requirement for these variables. For the remaining regression covariates, the global Schoenfeld test was nonsignificant ($\chi^2 = 3.958$, df = 8, $p = 0.861$). The IBC-status test was also nonsignificant ($\chi^2 = 0.025$, df = 1, $p = 0.874$; Figure S3).



(a) Non-IBC LABC



(b) Inflammatory Breast Cancer

Fig. 4: Kaplan–Meier estimates of breast cancer-specific survival (BCSS) stratified by race within (a) non-IBC locally advanced breast cancer (LABC) and (b) inflammatory breast cancer cohorts.

3.5 Competing Risks and Sensitivity Analysis

Cumulative incidence functions for breast cancer mortality in the presence of competing causes of death are shown in Figure 5. The cumulative incidence of breast cancer death was consistently higher among patients with IBC than among patients with non-IBC LABC. In contrast, the cumulative incidence of death from other causes was lower in the IBC group.

The Fine–Gray competing-risk model was used to evaluate whether the estimated association between IBC status and breast cancer mortality was sensitive to the treatment of deaths from other causes. The Fine–Gray model yielded a subdistribution HR of 1.377 (95% CI 1.163–1.631) for IBC compared with non-IBC LABC. This estimate was similar in direction and magnitude to the corresponding cause-specific Cox estimate of HR 1.415 (95% CI 1.196–1.675), as presented in Table S2. The cause-specific and subdistribution hazard estimates were similar in direction and magnitude, although the two models estimate different quantities and should be interpreted as complementary rather than interchangeable.

The competing-risk analysis did not evaluate whether the association between race and BCSS differed by inflammatory status. That question was evaluated using the primary race-by-IBC interaction model reported in Section 3.4. Unmeasured confounding, classification error, treatment timing, and selection arising from the complete-case restrictions remain possible.

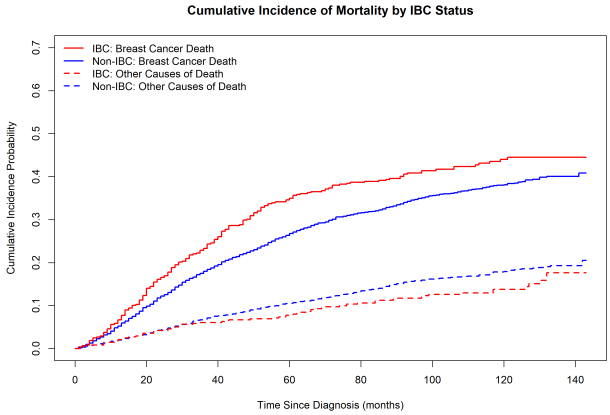


Fig. 5: Cumulative incidence functions (CIF) of breast cancer-specific mortality and other-cause mortality comparing inflammatory breast cancer (IBC) vs non-inflammatory locally advanced breast cancer (non-IBC LABC).

3.6 Propensity Score Sensitivity Analysis Results

Propensity-score matching yielded 50 Black and 50 White women in the IBC cohort and 305 Black and 305 White women in the non-IBC LABC cohort. The matched

non-IBC LABC cohort satisfied the specified balance criterion of an absolute standardized mean difference (SMD) below 0.10 for every included covariate, with a maximum absolute post-matching SMD of 0.074. In contrast, the matched IBC cohort did not satisfy the specified balance criterion because its maximum absolute post-matching SMD was 0.207. Complete covariate-balance results are presented in Figure S4 and Table S6.

Among the 100 matched IBC patients, including 46 breast cancer-specific deaths, the Cox model with robust standard errors yielded a Black-versus-White HR of 1.122 (95% CI 0.635–1.984, $p = 0.692$). This estimate is reported in Table S6 and was compatible with the primary adjusted IBC estimate of HR 1.033 (95% CI 0.632–1.689, $p = 0.897$) reported in Table 3. The corresponding unadjusted Kaplan–Meier comparison within the IBC cohort was also not statistically significant ($p = 0.160$; Figure 4b).

The matched IBC estimate remained imprecise, and the matched cohort retained residual measured imbalance. Accordingly, the matched IBC analysis was considered inconclusive rather than confirmatory and should receive limited inferential weight. It cannot establish whether the Black-versus-White association within IBC was absent, smaller, or larger than that observed within non-IBC LABC.

Among the 610 matched non-IBC LABC patients, including 243 breast cancer-specific deaths, Black women had a higher breast cancer-specific mortality hazard than White women (HR 1.369, 95% CI 1.063–1.764, $p = 0.015$). This estimate is presented in Table S6 and was directionally consistent with the primary adjusted non-IBC LABC estimate of HR 1.281 (95% CI 1.053–1.557, $p = 0.013$) reported in Table 3. The corresponding unadjusted survival patterns are shown in Figure 4a.

Overall, the matched non-IBC LABC analysis provided a more informative sensitivity analysis because its measured covariates satisfied the specified balance criterion, with a maximum absolute post-matching SMD of 0.074. In contrast, the matched IBC analysis remained inconclusive because its maximum absolute post-matching SMD was 0.207, the matched sample was small, and the confidence interval was wide. Propensity-score matching addressed measured baseline differences only and could not eliminate bias from unmeasured confounding. The matched analyses did not alter the inference from the formal race-by-IBC interaction analysis reported in Section 3.4. These matched estimates represent statistical associations and should not be interpreted as causal effects.

4 Discussion

In this population-based study of 3,174 women with AJCC 6th edition stage IIIB–IIIC breast cancer, IBC was associated with poorer BCSS than non-IBC LABC after adjustment for measured demographic, clinicopathologic, and treatment variables. The primary question, however, was whether the association between race and BCSS differed according to inflammatory status. In the common interaction model, the adjusted Black–White HR was 1.281 (95% CI: 1.053–1.557) in non-IBC LABC and 1.033 (95% CI: 0.632–1.689) in IBC. The Black-by-IBC interaction estimate, expressed as the IBC-to-non-IBC ratio of HRs, was 0.807 (95% CI: 0.477–1.365, $p = 0.423$). Neither the omnibus Wald test ($\chi^2(2) = 1.239$, $p = 0.538$) nor the likelihood-ratio test comparing models with and without the race-by-IBC interaction terms ($\chi^2(2) = 1.278$, $p = 0.528$) was statistically significant. Therefore, the study did not demonstrate that the race-BCSS association differed between IBC and non-IBC LABC. A statistically significant association in one subgroup and a nonsignificant association in another does not establish effect modification [42].

Precision is central to interpretation of the IBC estimate. The Black–White IBC comparison included only 54 Black women and 379 White women. The minimum detectable HR with 80% power was 1.856, whereas the approximate power to detect an HR of 1.281, corresponding to the observed non-IBC LABC association, was only 20.17% [44]. The 95% confidence interval for the Black-versus-White association within IBC was wide and included the null value as well as effect sizes

compatible with the non-IBC estimate. Similarly, the 95% confidence interval for the Black-by-IBC ratio of HRs crossed 1 and included values on both sides of the null interaction value. Consequently, neither the nonsignificant IBC subgroup estimate nor the nonsignificant interaction test establishes that the race association is absent, smaller, or larger within IBC. The results instead indicate substantial uncertainty regarding the magnitude of the race association within the IBC subgroup.

Previous studies have reported racial and ethnic survival differences among patients with IBC, but these studies addressed related rather than identical questions [32,36,37,38,39,40]. Schinkel et al. reported poorer survival among Black women in both inflammatory and non-inflammatory breast cancer, with Black-versus-White associations of broadly similar magnitude across inflammatory status [36]. Denu et al. reported more pronounced racial and socioeconomic disparities in IBC than in other breast cancers [37]. However, that conclusion is not directly equivalent to the present study's stage-restricted race-by-IBC interaction estimand. Differences in the comparator population, stage composition, outcome definition, covariate adjustment, socioeconomic information, and interaction-testing approach may contribute to the different interpretations. Other registry-based studies examined racial and ethnic survival differences primarily within IBC populations rather than formally testing whether the magnitude of the race-survival association differed between stage-comparable IBC and non-IBC LABC groups [32,38,39]. These distinctions are important because evidence of a racial survival disparity within an IBC cohort does not, by itself, establish that the magnitude of that disparity is greater in IBC than in non-IBC disease. Similarly, descriptive differences between subgroup-specific estimates do not estimate the same quantity as a formal race-by-inflammatory-status interaction. A recent systematic review and meta-analysis also reported poorer survival among Black patients with IBC while emphasizing substantial heterogeneity and residual confounding across observational studies [40].

These findings should not be interpreted as directly conflicting with the present analysis because the statistical questions, target populations, comparator groups, and estimands differ. The present study restricted both groups to AJCC stage IIIB–IIIC disease and formally tested race-by-inflammatory-status effect modification using interaction terms. Previous studies differed in geographic source population, diagnosis period, racial and ethnic classification, IBC case definition, stage eligibility, inclusion of metastatic disease, comparator population, survival endpoint, treatment availability, socioeconomic information, covariate adjustment, and statistical estimand. In addition, the relatively small Black IBC subgroup in the present cohort produced wide confidence intervals. Thus, the nonsignificant interaction in this study does not refute prior evidence of racial survival disparities among patients with IBC. Rather, this stage-restricted dataset did not provide sufficiently precise evidence to determine whether the magnitude of the race association differed by inflammatory status.

The propensity-score analysis provided different levels of information in the two inflammatory-status groups. In the matched non-IBC LABC sample, measured covariates satisfied the specified balance criterion, with a maximum absolute post-matching SMD of 0.074, and Black women had a higher breast cancer-specific mortality hazard than White women (HR 1.369, 95% CI 1.063–1.764). In contrast, the matched IBC sample retained residual measured imbalance, with a maximum absolute post-matching SMD of 0.207, and its Black-versus-White estimate remained imprecise (HR 1.122, 95% CI 0.635–1.984). Accordingly, the matched IBC analysis should receive limited inferential weight and cannot establish whether the Black-versus-White association within IBC was absent, smaller, or larger than that observed within non-IBC LABC. Propensity-score matching addressed measured baseline differences only and could not eliminate bias from unmeasured confounding [30,31].

The Fine–Gray subdistribution hazard estimate for IBC was similar in direction and magnitude to the cause-specific Cox estimate, and the cumulative incidence of

breast cancer death was higher among patients with IBC in the presence of competing mortality. Because cause-specific and subdistribution hazard models estimate different quantities, their similarity provides descriptive reassurance across modeling approaches but does not render the estimands interchangeable [19,20,21]. Neither analysis determines whether the race–survival association differs by inflammatory status. The AFT analyses similarly supported shorter survival among patients with IBC across the evaluated parametric distributions, but these models do not resolve the limited precision of the race-by-IBC comparison.

The observational data do not identify the mechanisms responsible for the estimated race associations. Potential pathways include tumor characteristics, treatment receipt and completion, treatment delays, comorbidity, insurance coverage, facility characteristics, socioeconomic conditions, healthcare access, and structural inequities [10,18,32,37]. SEER measures only a subset of these factors. Consequently, the present analyses cannot partition the observed associations into biological, treatment-related, healthcare, or social components. In particular, neither the subgroup estimates nor the race-by-IBC interaction test provides evidence that the aggressive biological phenotype of IBC itself attenuates racial survival disparities. Any mechanistic interpretation of differences between the subgroup point estimates would exceed what can be inferred from these observational registry data.

This study has several strengths. The stage-restricted design improved clinical comparability between IBC and non-IBC LABC and reduced confounding from comparisons across widely different stages. The analysis incorporated HR/HER2-defined molecular subtype, tumor grade, stage, recorded treatment, marital status, and age. The primary research question was evaluated using formal race-by-IBC interaction terms rather than comparisons of subgroup-specific p-values. Cause-specific Cox regression, Fine–Gray competing-risk regression, AFT models, propensity-score matching, proportional hazards diagnostics, and an event-based precision analysis were used to examine the robustness and limitations of the findings.

Several limitations require consideration. First, the number of Black women with IBC was small, resulting in wide confidence intervals and low power to detect an association comparable to that observed in non-IBC LABC. Second, SEER does not provide detailed information on chemotherapy regimens, dose intensity, endocrine therapy, HER2-directed treatment, treatment adherence, treatment completion, treatment timing, recurrence, performance status, insurance, facility characteristics, or detailed socioeconomic conditions. Residual confounding therefore remains likely. In addition, surgery, chemotherapy, and radiation therapy are post-diagnosis variables that may partly mediate associations of race and inflammatory status with survival. Therefore, models adjusted for these variables estimate conditional associations and should not be interpreted as estimating the total racial survival disparity.

Third, race was analyzed using the SEER race variable as White, Black, or API. Hispanic ethnicity was not modeled separately and was not used to define or exclude the analytic race groups. The results should therefore be interpreted as comparisons among the recorded SEER race categories rather than ethnicity-specific associations. The American Indian or Alaska Native group was excluded because its sample size was insufficient for stable group-specific estimation, limiting generalizability to that population.

Fourth, IBC classification relied on recorded AJCC T4d status and may be affected by registry coding or diagnostic misclassification. In addition, only two Grade IV tumors were available. Grades III and IV were pooled to avoid estimation based on an extremely sparse Grade IV category, but the two grades were not assumed to be prognostically equivalent. Pooling may obscure residual prognostic heterogeneity between poorly differentiated and undifferentiated disease and may leave residual confounding related to tumor differentiation. Nevertheless, exclusion of the two Grade IV observations in a sensitivity analysis did not materially alter the primary race-by-IBC interaction findings. Fifth, the stage re-

striction accounted for most exclusions from the source cohort. An additional 186 stage-eligible patients were excluded because of unknown molecular subtype, indeterminate surgery coding, or excluded or unknown race. No formal comparison of these excluded records with retained stage-eligible patients was conducted. The direction and magnitude of any selection bias arising from these exclusions cannot be determined.

Sixth, propensity-score matching addressed measured baseline characteristics only and cannot remove bias from unmeasured confounding. More importantly, the matched IBC sample did not satisfy the specified balance criterion, with a maximum absolute post-matching SMD of 0.207, and the resulting survival estimate remained imprecise. The IBC propensity-score result should therefore be interpreted as an inconclusive sensitivity analysis rather than as confirmatory evidence. The more favorable balance achieved in the non-IBC LABC matched sample does not resolve the limited information available for the IBC subgroup. Finally, the cohort included diagnoses from 2011 through 2015. Changes in HER2-directed treatment, immunotherapy, supportive care, and IBC management may limit the applicability of the estimates to patients treated in more recent periods.

Future studies should use larger multi-registry or multi-institutional cohorts with sufficient representation of Black patients with IBC. These studies should prespecify race-by-IBC interaction analyses and incorporate treatment timing, treatment completion, comorbidity, insurance, healthcare facility, socioeconomic, and neighborhood-level variables. More recent diagnosis periods may clarify whether race-associated survival patterns have changed with contemporary HER2-directed therapies, immunotherapy, and multidisciplinary IBC care. Greater subgroup size and precision are required before determining whether the magnitude of the race-survival association differs meaningfully between IBC and non-IBC LABC.

5 Conclusion

Among women with AJCC 6th edition stage IIIB-IIIC breast cancer, IBC was associated with poorer breast cancer-specific survival than non-IBC LABC. Black race was associated with higher breast cancer-specific mortality within non-IBC LABC, whereas the corresponding estimate within IBC was imprecise and not statistically significant. Formal race-by-IBC interaction tests did not provide statistically significant evidence that the race-BCSS association differed by inflammatory status. The Black-by-IBC interaction confidence interval included the null and values compatible with both smaller and larger race associations within IBC relative to non-IBC LABC. Accordingly, these data do not establish either attenuation or amplification of racial survival differences by inflammatory status.

The propensity-score analysis supported the non-IBC LABC finding but remained inconclusive within IBC because the matched IBC sample retained measured covariate imbalance, had a small sample size, and produced a wide confidence interval. All findings should be interpreted as associations derived from observational registry data. They do not establish that IBC biology causes or modifies racial survival disparities. Larger, clinically detailed cohorts with adequate representation of Black patients with IBC are needed to estimate race-by-inflammatory-status differences with greater precision and to investigate potential biological, treatment-related, healthcare, and social mechanisms.

Statements and Declarations

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Author contributions

Salah Uddin: Conceptualization, study design, data preparation, statistical analysis, interpretation of results, original manuscript drafting, and manuscript revision. Akhi Akter Chowdhury and Mahmudul Hasan: Manuscript editing. Haoxiang Feng: Manuscript review and editing. Md Monzur Murshed: Supervision, code verifica-

tion, statistical validation, and final approval of the manuscript.

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Data Availability

The data used in this study were obtained from the Surveillance, Epidemiology, and End Results (SEER) Program of the National Cancer Institute. De-identified SEER research data obtained under the SEER Research Data Agreement. The datasets analyzed in the current study are available from SEER at https://seer.cancer.gov.

Competing Interests

The authors declare no competing interests.

Ethical approval

This study used de-identified, publicly available SEER data. The study did not involve direct contact with human participants and did not include identifiable private information. Therefore, ethical approval and informed consent were not required.

Declaration of generative AI in scientific writing

While preparing this work, the authors used the generative pre-trained transformer models to check grammar. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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